UNIVERSITATEA TEHNICĂ DIN CLUJ-NAPOCA

Proiect: “Automat bancar”

Pentru: Proiectarea Sistemelor Numerice

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Cuprins

1. Cerința
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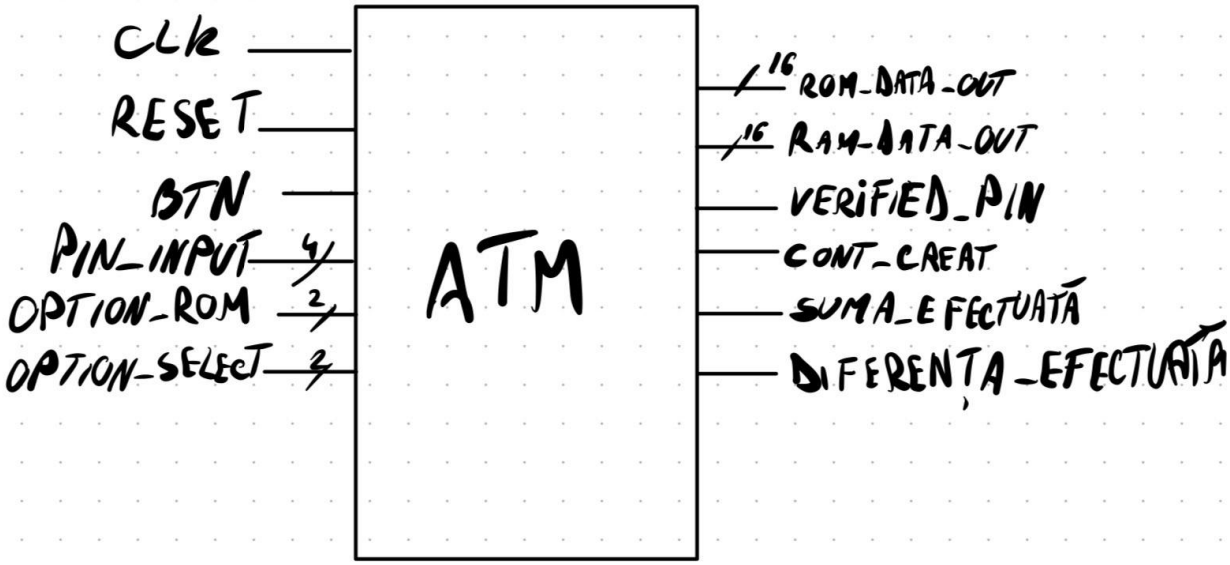
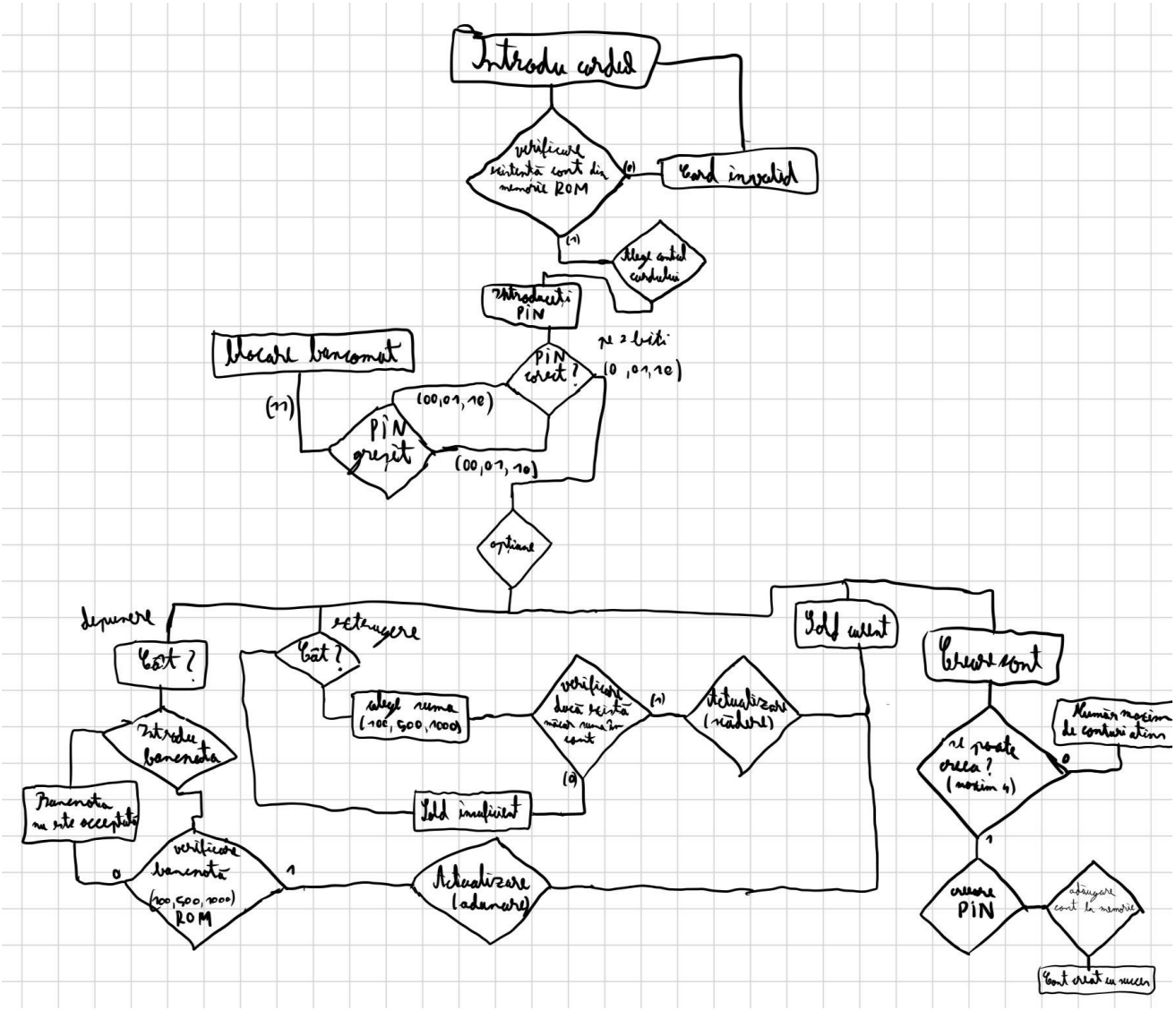
1. Cerința

Să se proiecteze un automat bancar pentru extrageri de sume in EURO. Se presupune că suma maxima care poate fi extras o data este de maximum 1.000 euro. Initial se efectueaza identificarea cardului si se alege operatia. Vor fi suportate minim 4 carduri/conturi diferite si se vor implementa minimum 4 operatii diferite. Automatul dispune de o casa in care initial se introduce o anumita sumă (număr de bancnote de diferite valori). În cazul cererii de eliberare de numerar se introduce suma, se verific existenta sumei cerute, se vizualizeaz tipurile de bancote emise si se actualizeaza contul.

2. Schema bloc și organigrama



Organigrama:



3. Componente:

ATM_Controller
Componenta principală care gestionează semnalele de control și interacțiunea cu celelalte module. Aceasta include un ROM pentru stocarea opțiunilor de operare, un RAM pentru stocarea conturilor și verificarea PIN-ului, un comparator pentru a verifica soldul disponibil, un subtractor pentru operațiunile de retragere și un sumator pentru operațiunile de depunere. Controlerul ATM asigură realizarea corectă a operațiunilor bancare prin intermediul acestor componente.

RAM_conturi
Modulul RAM stochează informațiile despre conturi și gestionează operațiunile de citire/scriere. Acesta are patru locații de memorie, fiecare putând stoca un sold diferit. Modulul permite citirea soldului curent și actualizarea acestuia în funcție de operațiunile de depunere sau retragere. De asemenea, verifică dacă PIN-ul introdus este corect și permite crearea unui nou cont dacă există spațiu disponibil.

Comparator_16bit
Comparatorul de 16 biți compară două valori de 16 biți și generează un semnal de ieșire dacă prima valoare este mai mică decât a doua. Aceasta este utilizată pentru a verifica dacă fondurile din cont sunt suficiente pentru o retragere. Comparatorul este esențial pentru a preveni efectuarea unor retrageri care ar duce la sold negativ.

Subtractor16Bit
Subtractorul de 16 biți scade două valori de 16 biți și generează diferența. Aceasta este utilizată pentru operațiunile de retragere de fonduri, asigurând calcularea corectă a soldului rămas după retragere. Subtractorul gestionează și împrumuturile necesare în cazul în care scăderea depășește valoarea inițială a soldului.

ROM
Modulul ROM furnizează date fixe pentru diferite operațiuni. Acesta este utilizat pentru stocarea opțiunilor de operare pentru controlerul ATM. ROM-ul stochează valorile fixe care sunt utilizate pentru selectarea diferitelor operațiuni bancare, cum ar fi retragerea sau depunerea unor sume prestabilite.

Sumator
Sumatorul adună două valori de 16 biți și generează suma acestora. Aceasta este utilizată pentru operațiunile de depunere de fonduri, asigurând adăugarea corectă a sumei depuse la soldul current. Sumatorul este esențial pentru gestionarea corectă a fondurilor în cadrul contului bancar.

4. Codul VHDL al componentelor:

atm_controller:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity ATM_Controller is
   Port (
      CLK : in STD_LOGIC;
      RESET : in STD_LOGIC;
      BTN : in STD_LOGIC; -- Added button input for debouncing
      PIN_Input : in STD_LOGIC_VECTOR(3 downto 0);
      Option_ROM: in STD_LOGIC_VECTOR(1 downto 0);
      Option_Select : in STD_LOGIC_VECTOR(1 downto 0);
      ROM_Data_Out : out STD_LOGIC_VECTOR(15 downto 0);--doar simulare
      RAM_Data_Out : out STD_LOGIC_VECTOR(15 downto 0);--afisam calcule pe parcurs
      Verified_PIN : out STD_LOGIC;--0 daca pin-ul este gresit si 1 daca este corect
      Cont_creat : out STD_LOGIC; -- Indicates if the account was successfully created (0 for success,
1 for failure)
      Suma_efectuata : out STD_LOGIC; --1 daca s-a facut suma 0 daca nu
      Diferenta_efectuata : out STD_LOGIC --1 daca s-a facut scaderea 0 daca nu
      );
end ATM_Controller;

architecture Behavioral of ATM_Controller is

   -- Debouncer component declaration
   component MPG is
      Port (
         btn : in STD_LOGIC;
         clk : in STD_LOGIC;
         en : out STD_LOGIC
      );
   end component;

   -- Other component declarations
   component comparator_16bit
      Port (
         A : in STD_LOGIC_VECTOR(15 downto 0);
         B : in STD_LOGIC_VECTOR(15 downto 0);
         AltB : out STD_LOGIC
      );
   end component;

   component Subtractor16Bit
      Port (
         A : in STD_LOGIC_VECTOR(15 downto 0);
         B : in STD_LOGIC_VECTOR(15 downto 0);
         Bin : in STD_LOGIC;
         Diff : out STD_LOGIC_VECTOR(15 downto 0);
         Bout : out STD_LOGIC
      );
   end component;
```

```
    component ROM
        Port (
            addresa : in STD_LOGIC_VECTOR(1 downto 0);
            data_out : out STD_LOGIC_VECTOR(15 downto 0)
        );
    end component;

    component ram_conturi
        Port (
            Adresa : in STD_LOGIC_VECTOR(3 downto 0);
            CS, RD, WR : in STD_LOGIC;
            D_In : in STD_LOGIC_VECTOR(15 downto 0);
            D_out : out STD_LOGIC_VECTOR(15 downto 0);
            pin_verificare : out STD_LOGIC;
            Cont_creat : out STD_LOGIC;
            Cont_creare: in std_logic
        );
    end component;

    component sumator
        Port (
            x : in STD_LOGIC_VECTOR(15 downto 0);
            y : in STD_LOGIC_VECTOR(15 downto 0);
            output : out STD_LOGIC_VECTOR(15 downto 0)
        );
    end component;

    -- Internal signals for inter-component communication
    signal internal_ROM_data : STD_LOGIC_VECTOR(15 downto 0);
    signal internal_RAM_data : STD_LOGIC_VECTOR(15 downto 0);
    signal internal_comparator_result : STD_LOGIC;
    signal internal_subtractor_diff : STD_LOGIC_VECTOR(15 downto 0);
    signal internal_sum_result : STD_LOGIC_VECTOR(15 downto 0);
    signal internal_verified_PIN : STD_LOGIC;
    signal internal_Cont_creat : STD_LOGIC;
    signal internal_borrow_out : STD_LOGIC;
    signal RD : STD_LOGIC;
    signal WR : STD_LOGIC;
    signal calcul_intermediar : STD_LOGIC_VECTOR(15 downto 0);
    signal Cont_creare: std_logic;

    -- Debouncer signals
    signal debounced_BTN : STD_LOGIC;

begin

    -- Debouncer instance
    debouncer_inst : MPG
        port map (
            btn => BTN,
            clk => CLK,
            en => debounced_BTN
        );

    -- ROM component instance
    rom_inst : ROM
```

```
    port map (
        addresa => Option_ROM,
        data_out => internal_ROM_data
    );

-- RAM component instance
ram_inst : ram_conturi
    port map (
        Adresa => PIN_Input,
        CS => '1',
        RD => RD,
        WR => WR,
        D_In => calcul_intermediar,
        D_out => internal_RAM_data,
        pin_verificare => internal_verified_PIN,
        Cont_creat => internal_Cont_creat,
        Cont_creare => Cont_creare
    );

-- Comparator component instance
comparator_inst : comparator_16bit
    port map (
        A => internal_ROM_data,
        B => internal_RAM_data,
        AltB => internal_comparator_result
    );

-- Subtractor component instance
subtractor_inst : Subtractor16Bit
    port map (
        A => internal_RAM_data,
        B => internal_ROM_data,
        Bin => '0',
        Diff => internal_subtractor_diff,
        Bout => internal_borrow_out
    );

-- Adder component instance
sumator_inst : sumator
    port map (
        x => internal_RAM_data,
        y => internal_ROM_data,
        output => internal_sum_result
    );

-- Main control logic
process (CLK, RESET)
begin

Cont_creat <= internal_Cont_creat;--am creat cont sau nu
Cont_creare <= '1';--vreau sa creez cont
RD <= '1';--verific mai intai daca pinul este corect
WR <= '0';

    if RESET = '1' then
        ROM_Data_Out <= (others => '0');
```

```
            RAM_Data_Out <= (others => '0');

        elsif rising_edge(CLK) then

        if internal_verified_PIN = '1' and debounced_BTN = '1' then --and debounced_BTN = '1' then --
Use debounced button signal

            Verified_PIN <= internal_verified_PIN;--pin bun sau nu

                case Option_Select is
                    when "00" => -- Account creation with an initial deposit of 10 euros
                    WR <= '1';
                    RD <= '0';
                    Cont_creare <= '1';--vreau sa creez cont
                    when "01" => -- Balance inquiry
                    RD <= '1';
                    WR <= '0';
                        RAM_Data_Out <= internal_RAM_data;
                    when "10" => -- Withdrawal
                        if internal_comparator_result = '1' then
                            RD <= '0';
                        WR <= '1';
                            calcul_intermediar <= internal_subtractor_diff;
                            RAM_Data_Out <= internal_subtractor_diff;--vreau sa vad valoarea
                        else
                            RAM_Data_Out <= (others => '0'); -- Insufficient funds
                        end if;
                    when "11" => -- Deposit
                        calcul_intermediar <= internal_sum_result;
                        RAM_Data_Out <= internal_sum_result;--vreau sa vad valoarea
                    when others =>
                        RAM_Data_Out <= (others => '0');
                        ROM_Data_Out <= (others => '0');
                end case;
            else
                RAM_Data_Out <= (others => '0');
            end if;
            ROM_Data_Out <= internal_ROM_data;
        end if;
    end process;

end Behavioral;
```

- RAM:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
```

```
--library UNISIM;
--use UNISIM.VComponents.all;

entity ram_conturi is
 Port (Adresa: in std_logic_vector(3 downto 0);--pin introdus de la tastatura pe biti direct
       CS, RD, WR : in std_logic;
       D_In: in std_logic_vector(15 downto 0);--suma bani intrare
       D_out: out std_logic_vector(15 downto 0);--sume bani iesire
       pin_verificare: out std_logic;--semnal daca pinul introdus a fost bun sau nu
       Cont_creat: out std_logic; -- semnal daca contul a fost creat
       Cont_creare: in std_logic --semnal daca vreau sa creez cont
       );

end ram_conturi;

architecture Behavioral of ram_conturi is
type MEM is array(0 to 3) of std_logic_vector(15 downto 0);
signal ROM: MEM :=
("0000000000110010","0000000100101100","0000101110111000","0000000000000000");--
50,300,3000,0
begin

   process(Adresa,CS,RD,WR)
   variable ok_pin:std_logic;
   variable Cont_creat_c:std_logic;
   begin

   Cont_creat_c :='1';
   ok_pin :='0';

   if CS = '0' then D_out <= "ZZZZZZZZZZZZZZZZ"; --daca nu e pornit RAM atunci afisam Z pe
iesire
   else
      if RD = '1' then -- daca citim
       case Adresa is
        when "1001" => D_out <= ROM(0);ok_pin := '1';
        when "1100" => D_out <= ROM(1);ok_pin := '1';
        when "1101" => D_out <= ROM(2);ok_pin := '1';
        when "1011" => D_out <= ROM(3);ok_pin := '1';
        when others => D_out <= "0000000000000000";--nu am nimerit nici un cont
           ok_pin := '0';
      end case;
      else if WR= '1' then --daca scriem

      if Cont_creare = '1' then
      for i in 0 to 3 loop

      if ROM(i) = "0000000000000000" then
      ROM(i) <= "0000000000000010";--Am creat contul si am adaugat 10 euro
      Cont_creat_c := '0';
      end if;

      end loop;
      end if;

      case Adresa is
```

```
        when "1001" => ROM(0) <= D_In;ok_pin := '1';
        when "1100" => ROM(1) <= D_In;ok_pin := '1';
        when "1101" => ROM(2) <= D_In;ok_pin := '1';
        when "1011" => ROM(3) <= D_In;ok_pin := '1';
        when others => D_out <= "0000000000000000";
        end case;
          else D_out <= "0000000000000000"; -- daca nu scriu si nu citesc
                ok_pin := '0';
          end if;
         end if;
       end if;
        pin_verificare <= ok_pin;
        Cont_creat <= Cont_creat_c;
        end process;
end Behavioral;
```

- ROM:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity ROM is
   Port (
      addresa : in STD_LOGIC_VECTOR(1 downto 0);  -- Adresa de 2 bi?i pentru 4 loca?ii (00, 01,
10, 11)
      data_out : out STD_LOGIC_VECTOR(15 downto 0)  -- Ie?irea datelor de 16 bi?i
   );
end ROM;
architecture Behavioral of ROM is
   type rom_array is array (0 to 3) of STD_LOGIC_VECTOR(15 downto 0);  -- ROM cu 4 loca?ii de
16 bi?i
   constant ROM_CONTENTS : rom_array := (
      "0000000001100100",  -- 100
      "0000000011111010",  -- 250
      "0000000111110100",  -- 500
      "0000001111101000"   --1000
   );
begin
   process(addresa)
   begin
      case addresa is
         when "00" =>
            data_out <= ROM_CONTENTS(0);
         when "01" =>
            data_out <= ROM_CONTENTS(1);
         when "10" =>
            data_out <= ROM_CONTENTS(2);
         when "11" =>
            data_out <= ROM_CONTENTS(3);
         when others =>
            data_out <= (others => '0');  -- Valoare implicit? pentru adrese invalide
      end case;
   end process;
end Behavioral;
```

- Sumator:

```
library IEEE;
```

```
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity sumator is
   Port ( x : in STD_LOGIC_VECTOR (15 downto 0);
          y : in STD_LOGIC_VECTOR (15 downto 0);
          output : out STD_LOGIC_VECTOR (15 downto 0));
end sumator;

architecture Behavioral of sumator is


begin
output <= x + y;

end Behavioral;
```

- Scăzător:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Subtractor16Bit is
   Port (
      A : in STD_LOGIC_VECTOR(15 downto 0);  -- Primul operand de 16 bi?i
      B : in STD_LOGIC_VECTOR(15 downto 0);  -- Al doilea operand de 16 bi?i
      Bin : in STD_LOGIC;  -- Împrumut de intrare (borrow in)
      Diff : out STD_LOGIC_VECTOR(15 downto 0);  -- Diferen?a de 16 bi?i
      Bout : out STD_LOGIC  -- Împrumut de ie?ire (borrow out)
   );
end Subtractor16Bit;

architecture Behavioral of Subtractor16Bit is
   -- Semnale pentru împrumuturile interne
   signal Borrow : STD_LOGIC_VECTOR(16 downto 0);

   -- Componenta sc?z?tor pe 1 bit
   component FullSubtractor1Bit
      Port (
         A : in STD_LOGIC;
         B : in STD_LOGIC;
         Bin : in STD_LOGIC;
         D : out STD_LOGIC;
         Bout : out STD_LOGIC
      );
   end component;

begin
```

```
    -- Instan?ierea sc?z?toarelor pe 1 bit ?i conectarea acestora
    gen_subtractors: for i in 0 to 15 generate
        bit_sub: FullSubtractor1Bit
            port map (
                A => A(i),
                B => B(i),
                Bin => Borrow(i),
                D => Diff(i),
                Bout => Borrow(i+1)
            );
    end generate;

    -- Setarea împrumutului ini?ial ?i final
    Borrow(0) <= Bin;
    Bout <= Borrow(16);

end Behavioral;
```

- Scăzător un bit:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity FullSubtractor1Bit is
    Port (
        A : in STD_LOGIC;  -- Primul operand (bitul A)
        B : in STD_LOGIC;  -- Al doilea operand (bitul B)
        Bin : in STD_LOGIC;  -- Împrumut de intrare (borrow in)
        D : out STD_LOGIC;  -- Diferen?a (A - B - Bin)
        Bout : out STD_LOGIC  -- Împrumut de ie?ire (borrow out)
    );
end FullSubtractor1Bit;

architecture Behavioral of FullSubtractor1Bit is
begin
    process(A, B, Bin)
    begin
        -- Calculul diferen?ei
        D <= A xor B xor Bin;

        -- Calculul împrumutului
        Bout <= (not A and B) or (not A and Bin) or (B and Bin);
    end process;
end Behavioral;
```

- Comparator:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity comparator_16bit is
    Port ( A : in  STD_LOGIC_VECTOR (15 downto 0);
           B : in  STD_LOGIC_VECTOR (15 downto 0);
           AltB : out  STD_LOGIC);
end comparator_16bit;

architecture Behavioral of comparator_16bit is
    signal bit_compare : STD_LOGIC_VECTOR (15 downto 0);
begin
```

```
    -- Instantiate 16 instances of the 1-bit comparator
    gen_comp: for i in 0 to 15 generate
        comp_inst: entity work.comparator_1bit
            port map (
                A => A(i),
                B => B(i),
                AltB => bit_compare(i)
            );
    end generate;

    process (bit_compare)
    begin
        -- Initialize AltB to '0'
        AltB <= '0';

        -- Check each bit from MSB to LSB
        for i in 15 downto 0 loop
            if (bit_compare(i) = '1') then
                AltB <= '1';
                exit;
            elsif (A(i) /= B(i)) then
                exit;
            end if;
        end loop;
    end process;

end Behavioral;
```

- Comparator pe un bit:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity comparator_1bit is
    Port ( A : in  STD_LOGIC;
           B : in  STD_LOGIC;
           AltB : out  STD_LOGIC);
end comparator_1bit;

architecture Behavioral of comparator_1bit is
begin
    process (A, B)
    begin
        if (A < B) then
            AltB <= '1';
        else
            AltB <= '0';
        end if;
    end process;
end Behavioral;
```

- Debouncer:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity MPG is
```

```
    Port ( btn : in STD_LOGIC;
           clk : in STD_LOGIC;
           en : out STD_LOGIC);
end MPG;
architecture Behavioral of MPG is
signal count : std_logic_vector (15 downto 0) := (others => '0');
signal t : std_logic;
signal q1, q2, q3 : std_logic;
begin
process (clk)
begin
    if (rising_edge(clk)) then
        count <= count + 1;
    end if;
end process;
t <= '1' when count = x"FFFF" else '0';
process(clk)
begin
    if (rising_edge(clk)) then
        if(t = '1') then
            q1 <= btn;
        end if;
    end if;
end process;
process(clk)
begin
    if (rising_edge(clk)) then
        q2 <= q1;
       q3 <= q2;
    end if;
end process;
en <= q2 and not q3;
end Behavioral;
```

Test bench pentru proiect:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity TB_ATM_Controller is
-- Testbench does not have any ports
end TB_ATM_Controller;

architecture Behavioral of TB_ATM_Controller is

    -- Component Declaration for the Unit Under Test (UUT)
    component ATM_Controller
        Port (
            CLK : in STD_LOGIC;
            RESET : in STD_LOGIC;
            BTN : in STD_LOGIC;
            PIN_Input : in STD_LOGIC_VECTOR(3 downto 0);
            Option_ROM: in STD_LOGIC_VECTOR(1 downto 0);
            Option_Select : in STD_LOGIC_VECTOR(1 downto 0);
            ROM_Data_Out : out STD_LOGIC_VECTOR(15 downto 0);
            RAM_Data_Out : out STD_LOGIC_VECTOR(15 downto 0);
            Verified_PIN : out STD_LOGIC;
```

```
        Cont_creat : out STD_LOGIC
    );
end component;

-- Signals to connect to the UUT
signal CLK : STD_LOGIC := '0';
signal RESET : STD_LOGIC := '0';
signal BTN : STD_LOGIC := '0';
signal PIN_Input : STD_LOGIC_VECTOR(3 downto 0) := (others => '0');
signal Option_ROM : STD_LOGIC_VECTOR(1 downto 0) := (others => '0');
signal Option_Select : STD_LOGIC_VECTOR(1 downto 0) := (others => '0');
signal ROM_Data_Out : STD_LOGIC_VECTOR(15 downto 0);
signal RAM_Data_Out : STD_LOGIC_VECTOR(15 downto 0);
signal Verified_PIN : STD_LOGIC;
signal Cont_creat : STD_LOGIC;

-- Clock period definition
constant CLK_PERIOD : time := 10 ns;

begin

-- Instantiate the Unit Under Test (UUT)
uut: ATM_Controller
    Port map (
        CLK => CLK,
        RESET => RESET,
        BTN => BTN,
        PIN_Input => PIN_Input,
        Option_ROM => Option_ROM,
        Option_Select => Option_Select,
        ROM_Data_Out => ROM_Data_Out,
        RAM_Data_Out => RAM_Data_Out,
        Verified_PIN => Verified_PIN,
        Cont_creat => Cont_creat
    );

-- Clock process definitions
CLK_process :process
begin
    CLK <= '0';
    wait for CLK_PERIOD/2;
    CLK <= '1';
    wait for CLK_PERIOD/2;
end process;

-- Stimulus process
stim_proc: process
begin
    -- Reset the ATM controller
    RESET <= '1';
    wait for CLK_PERIOD;
    RESET <= '0';

    -- Simulate account creation
    BTN <= '1';
    PIN_Input <= "1101"; -- Example PIN
```

```
        Option_ROM <= "00";
        wait for CLK_PERIOD * 10;
        assert (Verified_PIN = '1') report "Pin invalid" severity error;

        Option_Select <= "00"; -- Account creation
        wait for CLK_PERIOD * 10;
        assert (Cont_creat = '0') report "Nu s-a putut crea contul" severity error;
        BTN <= '0';

        -- Wait for some time
        wait for CLK_PERIOD * 20;

        -- Simulate balance inquiry
        BTN <= '1';
        Option_Select <= "01"; -- Balance inquiry
        wait for CLK_PERIOD * 10;
        BTN <= '0';

        -- Wait for some time
        wait for CLK_PERIOD * 20;

        -- Simulate deposit
        BTN <= '1';
        Option_Select <= "11"; -- Deposit
        wait for CLK_PERIOD * 10;
        BTN <= '0';

        -- Wait for some time
        wait for CLK_PERIOD * 20;

        -- Simulate withdrawal
        BTN <= '1';
        Option_Select <= "10"; -- Withdrawal
        wait for CLK_PERIOD * 10;
        BTN <= '0';

        -- Wait for some time
        wait for CLK_PERIOD * 20;
        assert (RAM_Data_Out = "0000000000110010") report "Nu s-a putut face retragerea" severity
error;
        -- Simulation end
        wait;
    end process;

end Behavioral;
```

Constraints Nexys 7:

```
## This file is a general .xdc for the Nexys A7-100T
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top level signal names in the
project

## Clock signal
set_property -dict { PACKAGE_PIN E3    IOSTANDARD LVCMOS33 } [get_ports { CLK100MHZ
}]; #IO_L12P_T1_MRCC_35 Sch=clk100mhz
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports {CLK100MHZ}];
```

```
##Switches
set_property -dict { PACKAGE_PIN J15   IOSTANDARD LVCMOS33 } [get_ports { RESET }];
#IO_L24N_T3_RS0_15 Sch=sw[0]
#set_property -dict { PACKAGE_PIN L16   IOSTANDARD LVCMOS33 } [get_ports { SW[1] }];
#IO_L3N_T0_DQS_EMCCLK_14 Sch=sw[1]
set_property -dict { PACKAGE_PIN M13   IOSTANDARD LVCMOS33 } [get_ports {
Option_ROM[0] }]; #IO_L6N_T0_D08_VREF_14 Sch=sw[2]
set_property -dict { PACKAGE_PIN R15   IOSTANDARD LVCMOS33 } [get_ports {
Option_ROM[1] }]; #IO_L13N_T2_MRCC_14 Sch=sw[3]
#set_property -dict { PACKAGE_PIN R17   IOSTANDARD LVCMOS33 } [get_ports { SW[4] }];
#IO_L12N_T1_MRCC_14 Sch=sw[4]
set_property -dict { PACKAGE_PIN T18   IOSTANDARD LVCMOS33 } [get_ports {
Option_Select[0] }]; #IO_L7N_T1_D10_14 Sch=sw[5]
set_property -dict { PACKAGE_PIN U18   IOSTANDARD LVCMOS33 } [get_ports {
Option_Select[1] }]; #IO_L17N_T2_A13_D29_14 Sch=sw[6]
#set_property -dict { PACKAGE_PIN R13   IOSTANDARD LVCMOS33 } [get_ports { SW[7] }];
#IO_L5N_T0_D07_14 Sch=sw[7]
set_property -dict { PACKAGE_PIN T8    IOSTANDARD LVCMOS18 } [get_ports { PIN_Input[0]
}]; #IO_L24N_T3_34 Sch=sw[8]
set_property -dict { PACKAGE_PIN U8    IOSTANDARD LVCMOS18 } [get_ports { PIN_Input[1]
}]; #IO_25_34 Sch=sw[9]
set_property -dict { PACKAGE_PIN R16   IOSTANDARD LVCMOS33 } [get_ports { PIN_Input[2]
}]; #IO_L15P_T2_DQS_RDWR_B_14 Sch=sw[10]
set_property -dict { PACKAGE_PIN T13   IOSTANDARD LVCMOS33 } [get_ports { PIN_Input[3]
}]; #IO_L23P_T3_A03_D19_14 Sch=sw[11]
#set_property -dict { PACKAGE_PIN H6    IOSTANDARD LVCMOS33 } [get_ports { SW[12] }];
#IO_L24P_T3_35 Sch=sw[12]
#set_property -dict { PACKAGE_PIN U12   IOSTANDARD LVCMOS33 } [get_ports { SW[13] }];
#IO_L20P_T3_A08_D24_14 Sch=sw[13]
#set_property -dict { PACKAGE_PIN U11   IOSTANDARD LVCMOS33 } [get_ports { SW[14] }];
#IO_L19N_T3_A09_D25_VREF_14 Sch=sw[14]
#set_property -dict { PACKAGE_PIN V10   IOSTANDARD LVCMOS33 } [get_ports { SW[15] }];
#IO_L21P_T3_DQS_14 Sch=sw[15]

## LEDs
set_property -dict { PACKAGE_PIN H17   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[0] }]; #IO_L18P_T2_A24_15 Sch=led[0]
set_property -dict { PACKAGE_PIN K15   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[1] }]; #IO_L24P_T3_RS1_15 Sch=led[1]
set_property -dict { PACKAGE_PIN J13   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[2] }]; #IO_L17N_T2_A25_15 Sch=led[2]
set_property -dict { PACKAGE_PIN N14   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[3] }]; #IO_L8P_T1_D11_14 Sch=led[3]
set_property -dict { PACKAGE_PIN R18   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[4] }]; #IO_L7P_T1_D09_14 Sch=led[4]
set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[5] }]; #IO_L18N_T2_A11_D27_14 Sch=led[5]
set_property -dict { PACKAGE_PIN U17   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[6] }]; #IO_L17P_T2_A14_D30_14 Sch=led[6]
set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[7] }]; #IO_L18P_T2_A12_D28_14 Sch=led[7]
set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[8] }]; #IO_L16N_T2_A15_D31_14 Sch=led[8]
```

```
set_property -dict { PACKAGE_PIN T15   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[9] }]; #IO_L14N_T2_SRCC_14 Sch=led[9]
set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[10] }]; #IO_L22P_T3_A05_D21_14 Sch=led[10]
set_property -dict { PACKAGE_PIN T16   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[11] }]; #IO_L15N_T2_DQS_DOUT_CSO_B_14 Sch=led[11]
set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[12] }]; #IO_L16P_T2_CSI_B_14 Sch=led[12]
set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[13] }]; #IO_L22N_T3_A04_D20_14 Sch=led[13]
set_property -dict { PACKAGE_PIN V12   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[14] }]; #IO_L20N_T3_A07_D23_14 Sch=led[14]
set_property -dict { PACKAGE_PIN V11   IOSTANDARD LVCMOS33 } [get_ports {
RAM_Data_Out[15] }]; #IO_L21N_T3_DQS_A06_D22_14 Sch=led[15]

## RGB LEDs
#set_property -dict { PACKAGE_PIN R12   IOSTANDARD LVCMOS33 } [get_ports { LED16_B }];
#IO_L5P_T0_D06_14 Sch=led16_b
#set_property -dict { PACKAGE_PIN M16   IOSTANDARD LVCMOS33 } [get_ports { LED16_G
}]; #IO_L10P_T1_D14_14 Sch=led16_g
#set_property -dict { PACKAGE_PIN N15   IOSTANDARD LVCMOS33 } [get_ports { LED16_R }];
#IO_L11P_T1_SRCC_14 Sch=led16_r
#set_property -dict { PACKAGE_PIN G14   IOSTANDARD LVCMOS33 } [get_ports { LED17_B }];
#IO_L15N_T2_DQS_ADV_B_15 Sch=led17_b
#set_property -dict { PACKAGE_PIN R11   IOSTANDARD LVCMOS33 } [get_ports { LED17_G }];
#IO_0_14 Sch=led17_g
#set_property -dict { PACKAGE_PIN N16   IOSTANDARD LVCMOS33 } [get_ports { LED17_R }];
#IO_L11N_T1_SRCC_14 Sch=led17_r

##7 segment display
#set_property -dict { PACKAGE_PIN T10   IOSTANDARD LVCMOS33 } [get_ports { CA }];
#IO_L24N_T3_A00_D16_14 Sch=ca
#set_property -dict { PACKAGE_PIN R10   IOSTANDARD LVCMOS33 } [get_ports { CB }];
#IO_25_14 Sch=cb
#set_property -dict { PACKAGE_PIN K16   IOSTANDARD LVCMOS33 } [get_ports { CC }];
#IO_25_15 Sch=cc
#set_property -dict { PACKAGE_PIN K13   IOSTANDARD LVCMOS33 } [get_ports { CD }];
#IO_L17P_T2_A26_15 Sch=cd
#set_property -dict { PACKAGE_PIN P15   IOSTANDARD LVCMOS33 } [get_ports { CE }];
#IO_L13P_T2_MRCC_14 Sch=ce
#set_property -dict { PACKAGE_PIN T11   IOSTANDARD LVCMOS33 } [get_ports { CF }];
#IO_L19P_T3_A10_D26_14 Sch=cf
#set_property -dict { PACKAGE_PIN L18   IOSTANDARD LVCMOS33 } [get_ports { CG }];
#IO_L4P_T0_D04_14 Sch=cg
#set_property -dict { PACKAGE_PIN H15   IOSTANDARD LVCMOS33 } [get_ports { DP }];
#IO_L19N_T3_A21_VREF_15 Sch=dp
#set_property -dict { PACKAGE_PIN J17   IOSTANDARD LVCMOS33 } [get_ports { AN[0] }];
#IO_L23P_T3_FOE_B_15 Sch=an[0]
#set_property -dict { PACKAGE_PIN J18   IOSTANDARD LVCMOS33 } [get_ports { AN[1] }];
#IO_L23N_T3_FWE_B_15 Sch=an[1]
#set_property -dict { PACKAGE_PIN T9    IOSTANDARD LVCMOS33 } [get_ports { AN[2] }];
#IO_L24P_T3_A01_D17_14 Sch=an[2]
#set_property -dict { PACKAGE_PIN J14   IOSTANDARD LVCMOS33 } [get_ports { AN[3] }];
#IO_L19P_T3_A22_15 Sch=an[3]
#set_property -dict { PACKAGE_PIN P14   IOSTANDARD LVCMOS33 } [get_ports { AN[4] }];
#IO_L8N_T1_D12_14 Sch=an[4]
```

```
#set_property -dict { PACKAGE_PIN T14   IOSTANDARD LVCMOS33 } [get_ports { AN[5] }];
#IO_L14P_T2_SRCC_14 Sch=an[5]
#set_property -dict { PACKAGE_PIN K2    IOSTANDARD LVCMOS33 } [get_ports { AN[6] }];
#IO_L23P_T3_35 Sch=an[6]
#set_property -dict { PACKAGE_PIN U13   IOSTANDARD LVCMOS33 } [get_ports { AN[7] }];
#IO_L23N_T3_A02_D18_14 Sch=an[7]

##CPU Reset Button
set_property -dict { PACKAGE_PIN C12   IOSTANDARD LVCMOS33 } [get_ports {
CPU_RESETN }]; #IO_L3P_T0_DQS_AD1P_15 Sch=cpu_resetn

##Buttons
set_property -dict { PACKAGE_PIN N17   IOSTANDARD LVCMOS33 } [get_ports { BTN }];
#IO_L9P_T1_DQS_14 Sch=btnc
#set_property -dict { PACKAGE_PIN M18   IOSTANDARD LVCMOS33 } [get_ports { BTNU }];
#IO_L4N_T0_D05_14 Sch=btnu
#set_property -dict { PACKAGE_PIN P17   IOSTANDARD LVCMOS33 } [get_ports { BTNL }];
#IO_L12P_T1_MRCC_14 Sch=btnl
#set_property -dict { PACKAGE_PIN M17   IOSTANDARD LVCMOS33 } [get_ports { BTNR }];
#IO_L10N_T1_D15_14 Sch=btnr
#set_property -dict { PACKAGE_PIN P18   IOSTANDARD LVCMOS33 } [get_ports { BTND }];
#IO_L9N_T1_DQS_D13_14 Sch=btnd


##Pmod Headers
##Pmod Header JA
#set_property -dict { PACKAGE_PIN C17   IOSTANDARD LVCMOS33 } [get_ports { JA[1] }];
#IO_L20N_T3_A19_15 Sch=ja[1]
#set_property -dict { PACKAGE_PIN D18   IOSTANDARD LVCMOS33 } [get_ports { JA[2] }];
#IO_L21N_T3_DQS_A18_15 Sch=ja[2]
#set_property -dict { PACKAGE_PIN E18   IOSTANDARD LVCMOS33 } [get_ports { JA[3] }];
#IO_L21P_T3_DQS_15 Sch=ja[3]
#set_property -dict { PACKAGE_PIN G17   IOSTANDARD LVCMOS33 } [get_ports { JA[4] }];
#IO_L18N_T2_A23_15 Sch=ja[4]
#set_property -dict { PACKAGE_PIN D17   IOSTANDARD LVCMOS33 } [get_ports { JA[7] }];
#IO_L16N_T2_A27_15 Sch=ja[7]
#set_property -dict { PACKAGE_PIN E17   IOSTANDARD LVCMOS33 } [get_ports { JA[8] }];
#IO_L16P_T2_A28_15 Sch=ja[8]
#set_property -dict { PACKAGE_PIN F18   IOSTANDARD LVCMOS33 } [get_ports { JA[9] }];
#IO_L22N_T3_A16_15 Sch=ja[9]
#set_property -dict { PACKAGE_PIN G18   IOSTANDARD LVCMOS33 } [get_ports { JA[10] }];
#IO_L22P_T3_A17_15 Sch=ja[10]

##Pmod Header JB
#set_property -dict { PACKAGE_PIN D14   IOSTANDARD LVCMOS33 } [get_ports { JB[1] }];
#IO_L1P_T0_AD0P_15 Sch=jb[1]
#set_property -dict { PACKAGE_PIN F16   IOSTANDARD LVCMOS33 } [get_ports { JB[2] }];
#IO_L14N_T2_SRCC_15 Sch=jb[2]
#set_property -dict { PACKAGE_PIN G16   IOSTANDARD LVCMOS33 } [get_ports { JB[3] }];
#IO_L13N_T2_MRCC_15 Sch=jb[3]
#set_property -dict { PACKAGE_PIN H14   IOSTANDARD LVCMOS33 } [get_ports { JB[4] }];
#IO_L15P_T2_DQS_15 Sch=jb[4]
#set_property -dict { PACKAGE_PIN E16   IOSTANDARD LVCMOS33 } [get_ports { JB[7] }];
#IO_L11N_T1_SRCC_15 Sch=jb[7]
#set_property -dict { PACKAGE_PIN F13   IOSTANDARD LVCMOS33 } [get_ports { JB[8] }];
#IO_L5P_T0_AD9P_15 Sch=jb[8]
```

```
#set_property -dict { PACKAGE_PIN G13   IOSTANDARD LVCMOS33 } [get_ports { JB[9] }];
#IO_0_15 Sch=jb[9]
#set_property -dict { PACKAGE_PIN H16   IOSTANDARD LVCMOS33 } [get_ports { JB[10] }];
#IO_L13P_T2_MRCC_15 Sch=jb[10]

##Pmod Header JC
#set_property -dict { PACKAGE_PIN K1    IOSTANDARD LVCMOS33 } [get_ports { JC[1] }];
#IO_L23N_T3_35 Sch=jc[1]
#set_property -dict { PACKAGE_PIN F6    IOSTANDARD LVCMOS33 } [get_ports { JC[2] }];
#IO_L19N_T3_VREF_35 Sch=jc[2]
#set_property -dict { PACKAGE_PIN J2    IOSTANDARD LVCMOS33 } [get_ports { JC[3] }];
#IO_L22N_T3_35 Sch=jc[3]
#set_property -dict { PACKAGE_PIN G6    IOSTANDARD LVCMOS33 } [get_ports { JC[4] }];
#IO_L19P_T3_35 Sch=jc[4]
#set_property -dict { PACKAGE_PIN E7    IOSTANDARD LVCMOS33 } [get_ports { JC[7] }];
#IO_L6P_T0_35 Sch=jc[7]
#set_property -dict { PACKAGE_PIN J3    IOSTANDARD LVCMOS33 } [get_ports { JC[8] }];
#IO_L22P_T3_35 Sch=jc[8]
#set_property -dict { PACKAGE_PIN J4    IOSTANDARD LVCMOS33 } [get_ports { JC[9] }];
#IO_L21P_T3_DQS_35 Sch=jc[9]
#set_property -dict { PACKAGE_PIN E6    IOSTANDARD LVCMOS33 } [get_ports { JC[10] }];
#IO_L5P_T0_AD13P_35 Sch=jc[10]

##Pmod Header JD
#set_property -dict { PACKAGE_PIN H4    IOSTANDARD LVCMOS33 } [get_ports { JD[1] }];
#IO_L21N_T3_DQS_35 Sch=jd[1]
#set_property -dict { PACKAGE_PIN H1    IOSTANDARD LVCMOS33 } [get_ports { JD[2] }];
#IO_L17P_T2_35 Sch=jd[2]
#set_property -dict { PACKAGE_PIN G1    IOSTANDARD LVCMOS33 } [get_ports { JD[3] }];
#IO_L17N_T2_35 Sch=jd[3]
#set_property -dict { PACKAGE_PIN G3    IOSTANDARD LVCMOS33 } [get_ports { JD[4] }];
#IO_L20N_T3_35 Sch=jd[4]
#set_property -dict { PACKAGE_PIN H2    IOSTANDARD LVCMOS33 } [get_ports { JD[7] }];
#IO_L15P_T2_DQS_35 Sch=jd[7]
#set_property -dict { PACKAGE_PIN G4    IOSTANDARD LVCMOS33 } [get_ports { JD[8] }];
#IO_L20P_T3_35 Sch=jd[8]
#set_property -dict { PACKAGE_PIN G2    IOSTANDARD LVCMOS33 } [get_ports { JD[9] }];
#IO_L15N_T2_DQS_35 Sch=jd[9]
#set_property -dict { PACKAGE_PIN F3    IOSTANDARD LVCMOS33 } [get_ports { JD[10] }];
#IO_L13N_T2_MRCC_35 Sch=jd[10]

##Pmod Header JXADC
#set_property -dict { PACKAGE_PIN A14   IOSTANDARD LVCMOS33 } [get_ports { XA_N[1] }];
#IO_L9N_T1_DQS_AD3N_15 Sch=xa_n[1]
#set_property -dict { PACKAGE_PIN A13   IOSTANDARD LVCMOS33 } [get_ports { XA_P[1] }];
#IO_L9P_T1_DQS_AD3P_15 Sch=xa_p[1]
#set_property -dict { PACKAGE_PIN A16   IOSTANDARD LVCMOS33 } [get_ports { XA_N[2] }];
#IO_L8N_T1_AD10N_15 Sch=xa_n[2]
#set_property -dict { PACKAGE_PIN A15   IOSTANDARD LVCMOS33 } [get_ports { XA_P[2] }];
#IO_L8P_T1_AD10P_15 Sch=xa_p[2]
#set_property -dict { PACKAGE_PIN B17   IOSTANDARD LVCMOS33 } [get_ports { XA_N[3] }];
#IO_L7N_T1_AD2N_15 Sch=xa_n[3]
#set_property -dict { PACKAGE_PIN B16   IOSTANDARD LVCMOS33 } [get_ports { XA_P[3] }];
#IO_L7P_T1_AD2P_15 Sch=xa_p[3]
#set_property -dict { PACKAGE_PIN A18   IOSTANDARD LVCMOS33 } [get_ports { XA_N[4] }];
#IO_L10N_T1_AD11N_15 Sch=xa_n[4]
```

```
#set_property -dict { PACKAGE_PIN B18   IOSTANDARD LVCMOS33 } [get_ports { XA_P[4] }];
#IO_L10P_T1_AD11P_15 Sch=xa_p[4]

##VGA Connector
#set_property -dict { PACKAGE_PIN A3    IOSTANDARD LVCMOS33 } [get_ports { VGA_R[0] }];
#IO_L8N_T1_AD14N_35 Sch=vga_r[0]
#set_property -dict { PACKAGE_PIN B4    IOSTANDARD LVCMOS33 } [get_ports { VGA_R[1] }];
#IO_L7N_T1_AD6N_35 Sch=vga_r[1]
#set_property -dict { PACKAGE_PIN C5    IOSTANDARD LVCMOS33 } [get_ports { VGA_R[2] }];
#IO_L1N_T0_AD4N_35 Sch=vga_r[2]
#set_property -dict { PACKAGE_PIN A4    IOSTANDARD LVCMOS33 } [get_ports { VGA_R[3] }];
#IO_L8P_T1_AD14P_35 Sch=vga_r[3]
#set_property -dict { PACKAGE_PIN C6    IOSTANDARD LVCMOS33 } [get_ports { VGA_G[0] }];
#IO_L1P_T0_AD4P_35 Sch=vga_g[0]
#set_property -dict { PACKAGE_PIN A5    IOSTANDARD LVCMOS33 } [get_ports { VGA_G[1] }];
#IO_L3N_T0_DQS_AD5N_35 Sch=vga_g[1]
#set_property -dict { PACKAGE_PIN B6    IOSTANDARD LVCMOS33 } [get_ports { VGA_G[2] }];
#IO_L2N_T0_AD12N_35 Sch=vga_g[2]
#set_property -dict { PACKAGE_PIN A6    IOSTANDARD LVCMOS33 } [get_ports { VGA_G[3] }];
#IO_L3P_T0_DQS_AD5P_35 Sch=vga_g[3]
#set_property -dict { PACKAGE_PIN B7    IOSTANDARD LVCMOS33 } [get_ports { VGA_B[0] }];
#IO_L2P_T0_AD12P_35 Sch=vga_b[0]
#set_property -dict { PACKAGE_PIN C7    IOSTANDARD LVCMOS33 } [get_ports { VGA_B[1] }];
#IO_L4N_T0_35 Sch=vga_b[1]
#set_property -dict { PACKAGE_PIN D7    IOSTANDARD LVCMOS33 } [get_ports { VGA_B[2] }];
#IO_L6N_T0_VREF_35 Sch=vga_b[2]
#set_property -dict { PACKAGE_PIN D8    IOSTANDARD LVCMOS33 } [get_ports { VGA_B[3] }];
#IO_L4P_T0_35 Sch=vga_b[3]
#set_property -dict { PACKAGE_PIN B11   IOSTANDARD LVCMOS33 } [get_ports { VGA_HS }];
#IO_L4P_T0_15 Sch=vga_hs
#set_property -dict { PACKAGE_PIN B12   IOSTANDARD LVCMOS33 } [get_ports { VGA_VS }];
#IO_L3N_T0_DQS_AD1N_15 Sch=vga_vs

##Micro SD Connector
#set_property -dict { PACKAGE_PIN E2    IOSTANDARD LVCMOS33 } [get_ports { SD_RESET
}]; #IO_L14P_T2_SRCC_35 Sch=sd_reset
#set_property -dict { PACKAGE_PIN A1    IOSTANDARD LVCMOS33 } [get_ports { SD_CD }];
#IO_L9N_T1_DQS_AD7N_35 Sch=sd_cd
#set_property -dict { PACKAGE_PIN B1    IOSTANDARD LVCMOS33 } [get_ports { SD_SCK }];
#IO_L9P_T1_DQS_AD7P_35 Sch=sd_sck
#set_property -dict { PACKAGE_PIN C1    IOSTANDARD LVCMOS33 } [get_ports { SD_CMD }];
#IO_L16N_T2_35 Sch=sd_cmd
#set_property -dict { PACKAGE_PIN C2    IOSTANDARD LVCMOS33 } [get_ports { SD_DAT[0]
}]; #IO_L16P_T2_35 Sch=sd_dat[0]
#set_property -dict { PACKAGE_PIN E1    IOSTANDARD LVCMOS33 } [get_ports { SD_DAT[1]
}]; #IO_L18N_T2_35 Sch=sd_dat[1]
#set_property -dict { PACKAGE_PIN F1    IOSTANDARD LVCMOS33 } [get_ports { SD_DAT[2]
}]; #IO_L18P_T2_35 Sch=sd_dat[2]
#set_property -dict { PACKAGE_PIN D2    IOSTANDARD LVCMOS33 } [get_ports { SD_DAT[3]
}]; #IO_L14N_T2_SRCC_35 Sch=sd_dat[3]

##Accelerometer
#set_property -dict { PACKAGE_PIN E15   IOSTANDARD LVCMOS33 } [get_ports { ACL_MISO
}]; #IO_L11P_T1_SRCC_15 Sch=acl_miso
#set_property -dict { PACKAGE_PIN F14   IOSTANDARD LVCMOS33 } [get_ports { ACL_MOSI
}]; #IO_L5N_T0_AD9N_15 Sch=acl_mosi
```

```
#set_property -dict { PACKAGE_PIN F15   IOSTANDARD LVCMOS33 } [get_ports { ACL_SCLK
}]; #IO_L14P_T2_SRCC_15 Sch=acl_sclk
#set_property -dict { PACKAGE_PIN D15   IOSTANDARD LVCMOS33 } [get_ports { ACL_CSN
}]; #IO_L12P_T1_MRCC_15 Sch=acl_csn
#set_property -dict { PACKAGE_PIN B13   IOSTANDARD LVCMOS33 } [get_ports { ACL_INT[1]
}]; #IO_L2P_T0_AD8P_15 Sch=acl_int[1]
#set_property -dict { PACKAGE_PIN C16   IOSTANDARD LVCMOS33 } [get_ports { ACL_INT[2]
}]; #IO_L20P_T3_A20_15 Sch=acl_int[2]

##Temperature Sensor
#set_property -dict { PACKAGE_PIN C14   IOSTANDARD LVCMOS33 } [get_ports { TMP_SCL
}]; #IO_L1N_T0_AD0N_15 Sch=tmp_scl
#set_property -dict { PACKAGE_PIN C15   IOSTANDARD LVCMOS33 } [get_ports { TMP_SDA
}]; #IO_L12N_T1_MRCC_15 Sch=tmp_sda
#set_property -dict { PACKAGE_PIN D13   IOSTANDARD LVCMOS33 } [get_ports { TMP_INT }];
#IO_L6N_T0_VREF_15 Sch=tmp_int
#set_property -dict { PACKAGE_PIN B14   IOSTANDARD LVCMOS33 } [get_ports { TMP_CT }];
#IO_L2N_T0_AD8N_15 Sch=tmp_ct

##Omnidirectional Microphone
#set_property -dict { PACKAGE_PIN J5    IOSTANDARD LVCMOS33 } [get_ports { M_CLK }];
#IO_25_35 Sch=m_clk
#set_property -dict { PACKAGE_PIN H5    IOSTANDARD LVCMOS33 } [get_ports { M_DATA }];
#IO_L24N_T3_35 Sch=m_data
#set_property -dict { PACKAGE_PIN F5    IOSTANDARD LVCMOS33 } [get_ports { M_LRSEL }];
#IO_0_35 Sch=m_lrsel

##PWM Audio Amplifier
#set_property -dict { PACKAGE_PIN A11   IOSTANDARD LVCMOS33 } [get_ports { AUD_PWM
}]; #IO_L4N_T0_15 Sch=aud_pwm
#set_property -dict { PACKAGE_PIN D12   IOSTANDARD LVCMOS33 } [get_ports { AUD_SD }];
#IO_L6P_T0_15 Sch=aud_sd

##USB-RS232 Interface
#set_property -dict { PACKAGE_PIN C4    IOSTANDARD LVCMOS33 } [get_ports {
UART_TXD_IN }]; #IO_L7P_T1_AD6P_35 Sch=uart_txd_in
#set_property -dict { PACKAGE_PIN D4    IOSTANDARD LVCMOS33 } [get_ports {
UART_RXD_OUT }]; #IO_L11N_T1_SRCC_35 Sch=uart_rxd_out
#set_property -dict { PACKAGE_PIN D3    IOSTANDARD LVCMOS33 } [get_ports { UART_CTS
}]; #IO_L12N_T1_MRCC_35 Sch=uart_cts
#set_property -dict { PACKAGE_PIN E5    IOSTANDARD LVCMOS33 } [get_ports { UART_RTS
}]; #IO_L5N_T0_AD13N_35 Sch=uart_rts

##USB HID (PS/2)
#set_property -dict { PACKAGE_PIN F4    IOSTANDARD LVCMOS33 } [get_ports { PS2_CLK }];
#IO_L13P_T2_MRCC_35 Sch=ps2_clk
#set_property -dict { PACKAGE_PIN B2    IOSTANDARD LVCMOS33 } [get_ports { PS2_DATA
}]; #IO_L10N_T1_AD15N_35 Sch=ps2_data

##SMSC Ethernet PHY
#set_property -dict { PACKAGE_PIN C9    IOSTANDARD LVCMOS33 } [get_ports { ETH_MDC
}]; #IO_L11P_T1_SRCC_16 Sch=eth_mdc
#set_property -dict { PACKAGE_PIN A9    IOSTANDARD LVCMOS33 } [get_ports { ETH_MDIO
}]; #IO_L14N_T2_SRCC_16 Sch=eth_mdio
#set_property -dict { PACKAGE_PIN B3    IOSTANDARD LVCMOS33 } [get_ports { ETH_RSTN
}]; #IO_L10P_T1_AD15P_35 Sch=eth_rstn
```

```
#set_property -dict { PACKAGE_PIN D9    IOSTANDARD LVCMOS33 } [get_ports { ETH_CRSDV
}]; #IO_L6N_T0_VREF_16 Sch=eth_crsdv
#set_property -dict { PACKAGE_PIN C10   IOSTANDARD LVCMOS33 } [get_ports {
ETH_RXERR }]; #IO_L13N_T2_MRCC_16 Sch=eth_rxerr
#set_property -dict { PACKAGE_PIN C11   IOSTANDARD LVCMOS33 } [get_ports {
ETH_RXD[0] }]; #IO_L13P_T2_MRCC_16 Sch=eth_rxd[0]
#set_property -dict { PACKAGE_PIN D10   IOSTANDARD LVCMOS33 } [get_ports {
ETH_RXD[1] }]; #IO_L19N_T3_VREF_16 Sch=eth_rxd[1]
#set_property -dict { PACKAGE_PIN B9    IOSTANDARD LVCMOS33 } [get_ports { ETH_TXEN
}]; #IO_L11N_T1_SRCC_16 Sch=eth_txen
#set_property -dict { PACKAGE_PIN A10   IOSTANDARD LVCMOS33 } [get_ports { ETH_TXD[0]
}]; #IO_L14P_T2_SRCC_16 Sch=eth_txd[0]
#set_property -dict { PACKAGE_PIN A8    IOSTANDARD LVCMOS33 } [get_ports { ETH_TXD[1]
}]; #IO_L12N_T1_MRCC_16 Sch=eth_txd[1]
#set_property -dict { PACKAGE_PIN D5    IOSTANDARD LVCMOS33 } [get_ports {
ETH_REFCLK }]; #IO_L11P_T1_SRCC_35 Sch=eth_refclk
#set_property -dict { PACKAGE_PIN B8    IOSTANDARD LVCMOS33 } [get_ports { ETH_INTN
}]; #IO_L12P_T1_MRCC_16 Sch=eth_intn

##Quad SPI Flash
#set_property -dict { PACKAGE_PIN K17   IOSTANDARD LVCMOS33 } [get_ports { QSPI_DQ[0]
}]; #IO_L1P_T0_D00_MOSI_14 Sch=qspi_dq[0]
#set_property -dict { PACKAGE_PIN K18   IOSTANDARD LVCMOS33 } [get_ports { QSPI_DQ[1]
}]; #IO_L1N_T0_D01_DIN_14 Sch=qspi_dq[1]
#set_property -dict { PACKAGE_PIN L14   IOSTANDARD LVCMOS33 } [get_ports { QSPI_DQ[2]
}]; #IO_L2P_T0_D02_14 Sch=qspi_dq[2]
#set_property -dict { PACKAGE_PIN M14   IOSTANDARD LVCMOS33 } [get_ports { QSPI_DQ[3]
}]; #IO_L2N_T0_D03_14 Sch=qspi_dq[3]
#set_property -dict { PACKAGE_PIN L13   IOSTANDARD LVCMOS33 } [get_ports { QSPI_CSN
}]; #IO_L6P_T0_FCS_B_14 Sch=qspi_csn
```